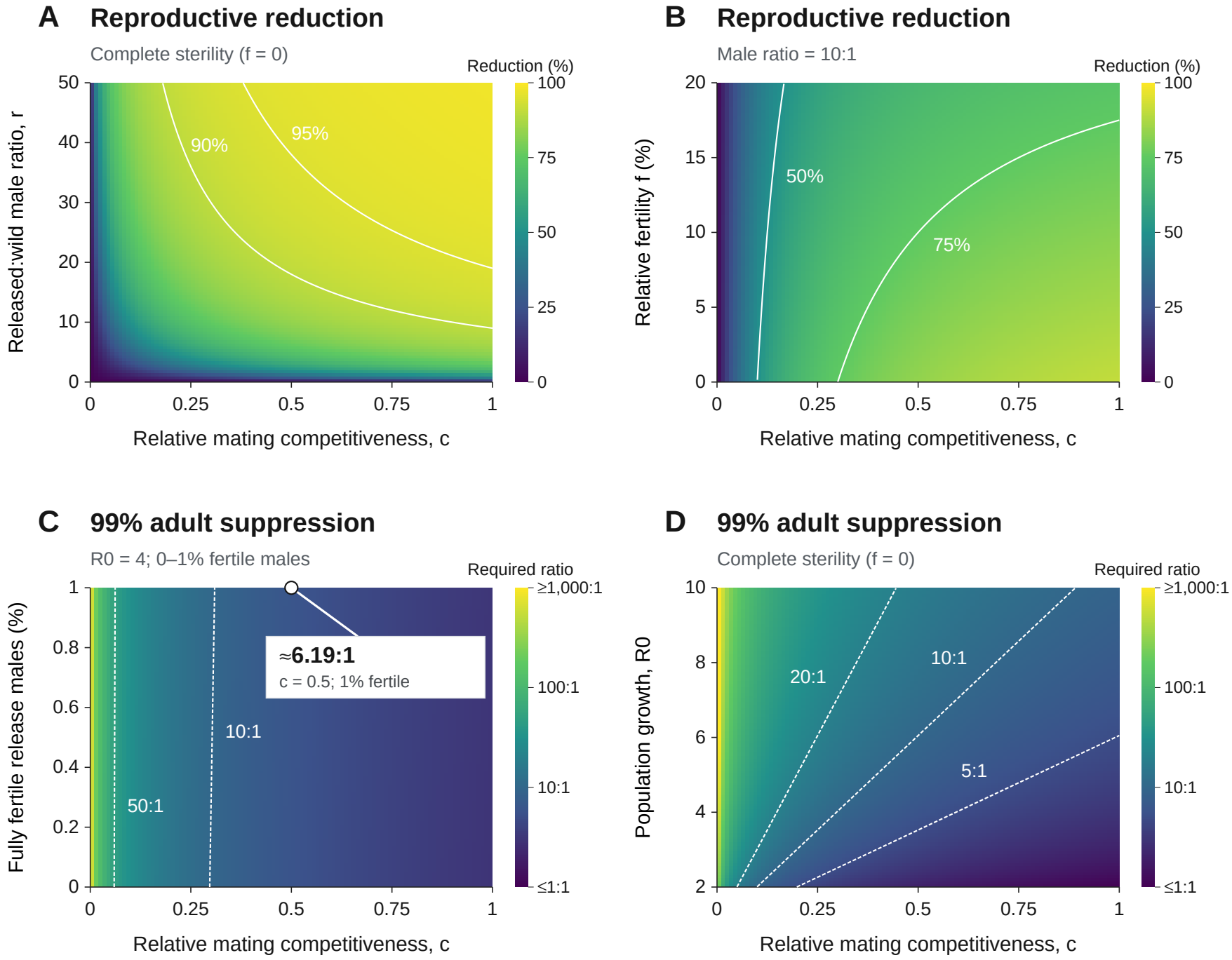

A
Reproductive reduction
Complete sterility (f = 0)
Reduction (%)
100
75
50
25
0
90%
95%
Released:wild male ratio, r
50
40
30
20
10
0
0
0.25
0.5
0.75
1
Relative mating competitiveness, c
B
Reproductive reduction
Male ratio = 10:1
Reduction (%)
100
75
50
25
0
50%
75%
Relative fertility f (%)
20
15
10
5
0
0
0.25
0.5
0.75
1
Relative mating competitiveness, c
C
99% adult suppression
R0 = 4; 0–1% fertile males
Required ratio
≥1,000:1
100:1
10:1
≤1:1
≈6.19:1
c = 0.5; 1% fertile
10:1
50:1
Fully fertile release males (%)
1
0.8
0.6
0.4
0.2
0
0
0.25
0.5
0.75
1
Relative mating competitiveness, c
D
99% adult suppression
Complete sterility (f = 0)
Required ratio
≥1,000:1
100:1
10:1
≤1:1
20:1
10:1
5:1
Population growth, R0
10
8
6
4
2
0
0.25
0.5
0.75
1
Relative mating competitiveness, c